



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME:

Sentinel



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1. Problem framing

Emergency triage is a two-minute call, made on thin data, while more patients keep arriving. It is not a clean ranking problem.

Age changes every number

A heart rate of 150 is an emergency in a calm adult and almost normal in a distressed toddler.

One adult-calibrated scale silently misreads children and the elderly.

Intake data is uneven

About half of arrivals have useful history on file; the other half are effectively strangers.

Tools that assume rich records fail worst on the people they know least.

Under-triage can kill

Over-triage only wastes some capacity, but missing a critical patient can be fatal.

Stroke, STEMI, and sepsis have treatment windows measured in minutes.

It does not end at the desk

Patients keep deteriorating in the waiting room, exactly when the department is busiest.

A score-once tool forgets them, so someone has to keep watching.

The real job is not average accuracy. It is to stay safe under sparse data and time pressure, keep watching the people who are waiting, and leave the final call to a clinician.

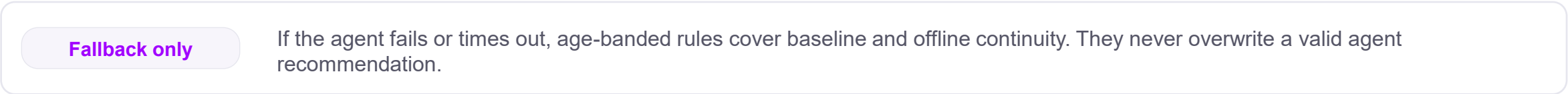
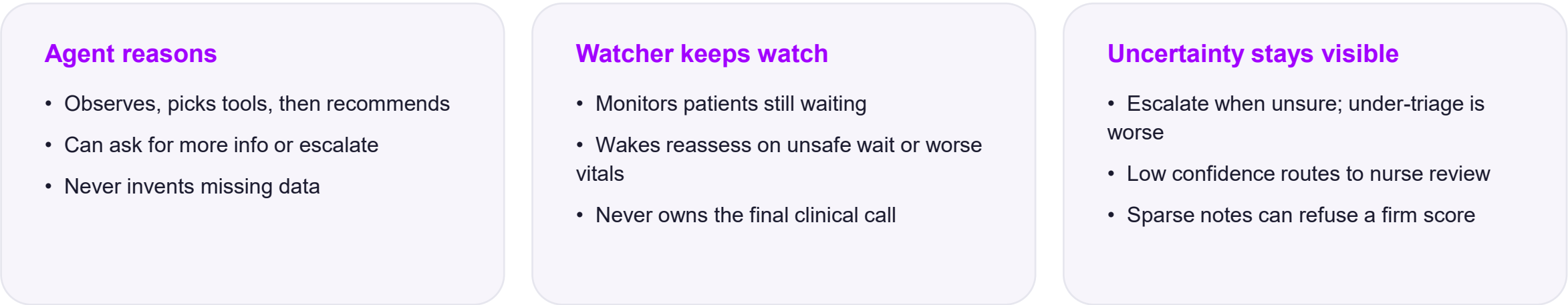
2. Solution overview

Sentinel recommends urgency and placement. A Gemini agent reasons with tools. A clinician makes the final call. No diagnosis, treatment, or discharge.

PRIMARY PATH



HOW IT STAYS SAFE



3. Data strategy and decision model

Built for incomplete intake, not the average chart. The Gemini agent decides; rules support, never silently override.

Data strategy

- Design for sparse arrivals. Missing fields lower confidence; they are never invented.
- When present: age, sex, arrival, complaint, pain, AVPU, vitals, onset, history, meds, allergies, prior-record flag.
- Weigh life-threat and age-banded danger-zone vitals first. Stroke, STEMI, and sepsis open clocks.
- About half the cohort has prior history, half does not. Zero-history walk-ins go to nurse review.
- Notes parse to structure. If the model is off or wrong, a rule parser keeps the desk moving.

Decision model

- Gemini agent is primary: tools, priority 1 to 5, confidence, and evidence.
- May request more information or escalate instead of guessing mid acuity.
- Rules are fallback only: baseline, offline continuity, auto-use on failure or timeout.
- A valid agent recommendation is never overwritten by the rules layer.
- Uncertainty stays visible. Thin mild notes can refuse a firm score. Eval compares agent, baseline, and clinician.

Invariants

Danger-zone vitals only push urgency up · Watcher never softens priority · Override / modify needs a reason · Original agent recommendation stays intact

4. Workflow and safety-first design

WORKFLOW

At the desk

The live board lists everyone waiting, most urgent first, with confidence and flags.

Open a patient to see evidence, gaps, and the baseline comparison.

Act in seconds: accept, modify, override, request more, or escalate.

Intake

A nurse pastes a short note with age, vitals, and a description.

One click parses it into structured fields and scores it.

Review the recommendation, or the needs-more gate, then add to the board.

Waiting room

The Watcher keeps scanning everyone who is still waiting.

It wakes a reassessment on unsafe wait time or worsening vitals.

It only escalates, and never softens priority because the floor got busy.

SAFETY RULES

Escalate when unsure

Upward-only ratchet

Clinician always final

Immutable audit trail

5. Prototype proof points

WHAT'S ALREADY BUILT

Phase 0, working in this repo on simulated data



25-patient synthetic board, most urgent first, live at the root URL.



Hard cases covered: ambiguous elder, febrile toddler, atypical geriatric MI, zero-history walk-in.



3x surge toggle plus a Watcher that catches a deteriorating waiter (P-011).



Confidence on every score. Low confidence routes to a nurse; thin notes can abstain.



Clinician override and modify with a required reason, on an append-only audit trail.



Age-banded vital thresholds, shown against the patient's own age band.

Posture

HIPAA (US) assumed. Non-Device Clinical Decision Support: it recommends, it does not command.

6. Target users

WHO USES IT

Triage nurse

The everyday user, working under real time pressure.

Gets a fast, explained starting point and clear confidence.

Accepts or adjusts in seconds without fighting the UI.

Charge nurse

Owens patient flow when the department is under load.

Sees who is waiting, who is overdue, and where surge is building.

The board and surge view are built for this role.

Emergency physician

Needs a defensible rationale behind each call.

Watches running clocks for time-critical syndromes.

Gets a clean record of every change and the reason for it.

Buyers

CMO and CNO for safety and liability. Operations for throughput and LWBS. CIO and CISO for integration and security.

7. Business case and impact

Assumed setting: US emergency departments, about 100 to 500+ visits a day, ESI-style acuity, HIPAA. Figures are illustrative, not a quote.

Safety PRIMARY	Cut under-triage of high-acuity patients. Age-aware vitals, escalate-when-unsure, and waiting-room monitoring attack the failure that matters most to a CMO. Even a small reduction in missed critical cases dominates the cost of the system.
Throughput HUNDREDS/YR	Faster, consistent first pass plus active re-check of the queue. At a mid-size site of about 40,000 visits a year, a one-point cut in leave-without-being-seen keeps hundreds of patients in care instead of walking out.
Time-critical MINUTES	Running clocks for stroke, STEMI, and sepsis surface treatable windows earlier. We do not invent site-specific minute savings; the product is built so those windows are hard to miss.
Cost to run Gemini is optional, cached, and iteration-capped. Quiet Watcher ticks skip model calls. With the model on, AI cost is a fraction of a cent per patient; with it off, the desk still runs on deterministic fallback.	Commercial model Per-department annual SaaS, tiered by visit volume. Pilots priced low or free in exchange for validation data and a reference site. The first credible outcomes study is worth more than year-one revenue.

8. Adoption, compliance, and scale

Adoption, privacy, and scale are treated as product work, not afterthoughts.

Adoption

- Shadow mode first: score beside triage, change nothing.
- Clinician in charge, always, with a reason on overrides.
- Fight alert fatigue: risk-weighted alerts, abstain on thin notes.
- Fit the existing two minutes, do not add clicks.

Compliance

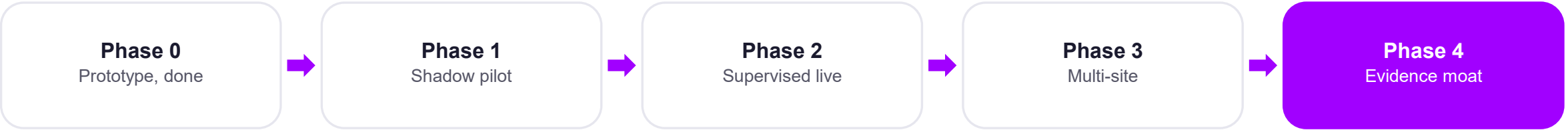
- HIPAA (US) assumed; audit and retention follow it.
- Minimize PHI to the model; BAA or on-prem option.
- Model-off mode loses none of the age-banded logic.
- Non-Device CDS: recommend, never command.

Scale

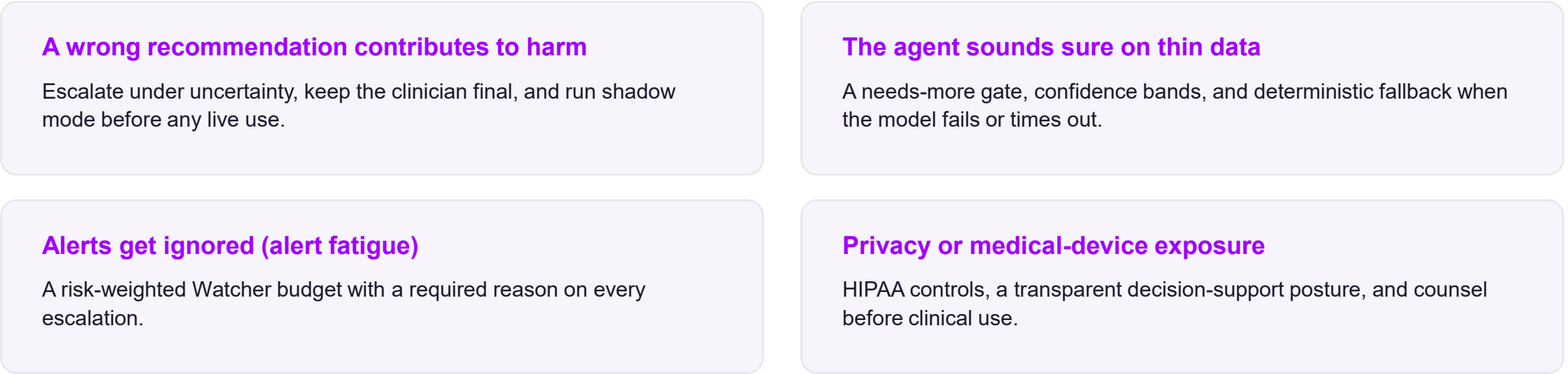
- Same core, different config, not forks.
- Age bands, safe-wait tables, and surge multipliers.
- Deterministic path is light enough for a hospital VM.
- Read-only shadow first, deeper write-back later.

9. Roadmap and risks

ROADMAP



RISKS AND MITIGATIONS



10. The ask

THE ASK

To turn a working prototype into a validated tool, we need three things.

1. Partner ED

One emergency department willing to run Sentinel in shadow mode, scoring beside real triage while changing no decisions at first.

2. Read-only access

Records and flow data over standard health-data interfaces, enough to score patients and measure agreement, nothing more.

3. Clinical champion

A clinician to help tune thresholds to the site, and to interpret where the assistant and the desk disagree.

In return

The site gets an early, safe view of its own mistriage and crowding, and a real say in how the assistant behaves.

Safe when it is unsure. Always watching the people who wait. The clinician always decides. That is the product Round 2 asked us to prove.

Thank you

